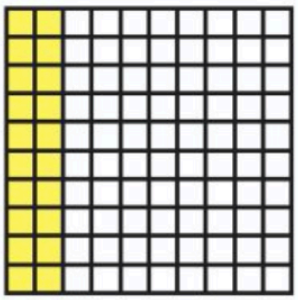
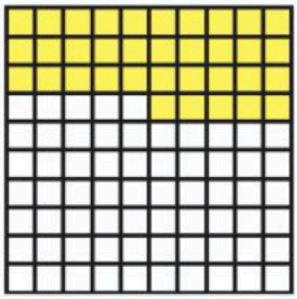
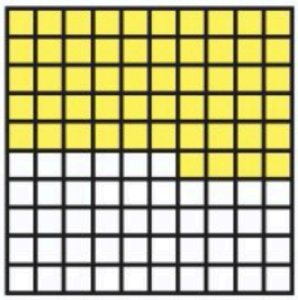
Exercise 7.1

1. Three figures, each with 100 squares, are shown below:

(a) 

(b) 

(c) 

(i) What percentage of each figure is coloured yellow?
(ii) What percentage of each figure is not coloured yellow?
(iii) What fraction of each figure is coloured yellow?
(iv) What decimal of each figure is coloured yellow?

2. Copy and complete each of these:

(i) $\frac{28}{100} = \square\%$ (ii) $\frac{54}{100} = \square\%$ (iii) $\frac{\square}{100} = 15\%$

(iv) $\frac{9}{10} = \frac{\square}{100} = \square\%$ (v) $\frac{11}{20} = \frac{\square}{100} = \square\%$ (vi) $\frac{14}{25} = \frac{\square}{100} = \square\%$

3. Express the following percentages as fractions in their simplest form:

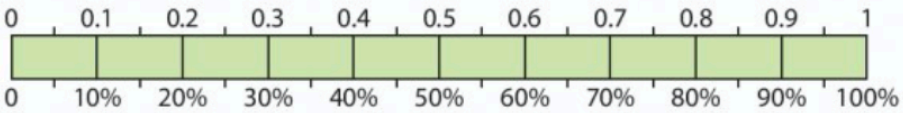
(i) 20% (ii) 60% (iii) 70% (iv) 15% (v) 25%
(vi) 75% (vii) 85% (viii) 5% (ix) 2% (x) 24%.

4. Copy and complete the following:

$$7\tfrac{1}{2}\% = \frac{7\frac{1}{2}}{100} = \frac{7\frac{1}{2} \times 2}{100 \times 2} = \frac{\square}{200} = \frac{3}{\square}$$

5. Change these percentages to fractions and simplify each fraction:

(i) $33\frac{1}{3}\%$ (ii) $12\frac{1}{2}\%$ (iii) $2\frac{1}{2}\%$ (iv) $8\frac{1}{3}\%$ (v) $17\frac{1}{2}\%$

6. 

Use the number line above to answer these questions:

(i) Write each of these decimals as a percentage:
(a) 0.5 (b) 0.2 (c) 0.35 (d) 0.8 (e) 0.75
(ii) Write each of these percentages as a decimal:
(a) 20% (b) 80% (c) 70% (d) 65% (e) 95%

7. Change these fractions to percentages:

(i) $\frac{18}{100}$ (ii) $\frac{47}{100}$ (iii) $\frac{3}{10}$ (iv) $\frac{7}{10}$ (v) $\frac{5}{20}$
(vi) $\frac{2}{5}$ (vii) $\frac{3}{4}$ (viii) $\frac{17}{20}$ (ix) $\frac{4}{25}$ (x) $\frac{19}{50}$

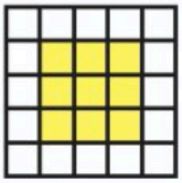
8. Write these percentages as decimals:

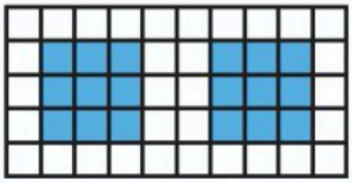
(i) 50% (ii) 10% (iii) 90% (iv) 35% (v) 45%
(vi) 8% (vii) $12\frac{1}{2}$% (viii) 24% (ix) 1% (x) $24\frac{1}{2}$%

9. Express the following decimals as percentages:

(i) 0.8 (ii) 0.4 (iii) 0.75 (iv) 0.45 (v) 0.15
(vi) 0.07 (vii) 0.01 (viii) 0.125 (ix) 0.345 (x) 0.045

10. What percentage of each diagram is shaded?

(i)

(ii)

(iii)


11. Put these in three matching groups.

$\frac{4}{5}$ 80% 0.8 40% $\frac{4}{10}$

$\frac{3}{4}$ $\frac{2}{5}$ 75% 0.75

12. Write the following quantities in order, starting with the smallest:

(i) 50% $\frac{1}{4}$ 20% 0.3 $\frac{2}{5}$
(ii) $\frac{3}{10}$ 60% $\frac{1}{5}$ $\frac{1}{2}$ $\frac{1}{4}$ 0.45

13. Write these fractions and percentages as decimals:

(i) $\frac{27}{100}$ (ii) $\frac{7}{100}$ (iii) 54% (iv) $17\frac{1}{2}$% (v) $2\frac{1}{2}$%

14. Find four matching pairs.
What could match the odd one out?

40% 0.25 15% 0.15 0.4

0.8 70% 80% 0.7

15.

A	E	U	N	R	S	T	M	V
$\frac{1}{2}$	30%	40%	0.8	0.6	0.2	75%	$\frac{1}{4}$	$\frac{7}{10}$

Use this code to find a letter for each fraction, decimal or percentage below.
Rearrange each set of letters to spell a planet.

(i) $\frac{3}{4}$ 0.5 $\frac{6}{10}$ 20% 80% $\frac{2}{5}$
(ii) 0.4 $\frac{1}{5}$ 0.7 $\frac{8}{10}$ 0.3
(iii) $\frac{3}{5}$ 50% 0.4 $\frac{1}{5}$ $\frac{4}{10}$ 80%